PRODUCT USER MANUAL

For NWS Ocean Waves Reanalysis Product

NWSHELF_REANALYSIS_WAV_004_015

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RECORD TABLE

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1.1	02/02/2021	All	added time series extension	M. Tonani, A. Saulter	I. Ascione
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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
NWS	North-West Shelf
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
PC	Production Centre
PU	Production Unit
DirectGetFile - DGF	CMEMS service tool (FTP like) to download a NetCDF file
WW-III	WaveWatch

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1 Summary

This document is the user manual for the CMEMS North West European Shelf reanalysis product **NWSHELF_REANALYSIS_ WAV_004_015**.

It provides reanalyses from 01/01/1980, updated twice a year.

The wave products are the integrated parameters computed from the total wave spectrum 3-hourly surface fields.

It is updated as follows twice a year:

- the time series is updated adding 6 months

The product is published on the CMEMS dissemination server after automatic and human quality controls. Product is available on-line and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.6 convention.

The reanalysis system is described in the Quality Information Document (QUID): http://marine.copernicus.eu/documents/QUID/CMEMS-NWS-QUID-004-015.pdf.

More detailed information can be obtained from the CMEMS Service Desk (servicedesk.cmems@mercator-ocean.eu).

II PRODUCT DESCRIPTION: NWSHELF_REANALYSIS_WAV _004_015

II.1 General Information about products

Product name	NWSHELF_REANALYSIS_WAV _004_015	
Geographical coverage	16°W → 13°E ; 46°N → 63°N	
Variables	Spectral significant wave height (Hm0) Spectral moments (-1,0) wave period (Tm-10) Spectral moments (0,2) wave period (Tm02) Wave period at spectral peak / peak period (Tp) Mean wave direction from (Mdir) Wave principal direction at spectral peak Stokes drift U Stokes drift V Spectral significant wind wave height Spectral moments (0,1) wind wave period Mean wind wave direction from Spectral significant primary swell wave height Spectral moments (0,1) primary swell wave period Mean primary swell wave direction from Spectral significant secondary swell wave height Spectral moments (0,1) secondary swell wave period Mean secondary swell wave direction from	
Update frequency	twice a year	
Available time series	From 01/01/1980 and regularly updated (see product improvements pages http://marine.copernicus.eu/services-portfolio/product-improvements/)	
Target delivery time	**N/A**	
Temporal resolution	3-hourly instantaneous	
Delivery mechanisms	Marine Data Store	
Horizontal resolution	1.9± 0.4 km longitude x 1.5km latitude	

	1/33° longitude x 1/74° latitude
Number of vertical levels	surface only
Format	NetCDF 4

II.2 Details of the datasets

NWSHELF_REANALYSIS_WAV _004_015	
MetO-NWS-WAV-RAN	**Variable name** in the NetCDF files (unit) Long_name Standard_name
	VHM0 [m] Spectral significant wave height (Hm0) sea_surface_wave_significant_height
	VTM10 [s] Spectral moments (-1,0) wave period (Tm-10) sea_surface_wave_mean_period_from_variance_spectral_density_inverse_frequency_moment
	VTM02 [s] Spectral moments (0,2) wave period (Tm02) sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_moment
	VTPK [s] Wave period at spectral peak / peak period (Tp) sea_surface_wave_period_at_variance_spectral_density_maximum
	VMDR [degree] Mean wave direction from (Mdir) sea_surface_wave_from_direction
	VPED [degree] Wave principal direction at spectral peak sea_surface_wave_from_direction_at_variance_spectral_density_maximum
	VSDX [m s-1] Stokes drift U sea_surface_wave_stokes_drift_x_velocity
	VSDY [m s-1] Stokes drift V sea_surface_wave_stokes_drift_y_velocity
	VHM0_WW [m] Spectral significant wind wave height sea_surface_wind_wave_significant_height
	VTM01_WW [s] Spectral moments (0,1) wind wave period sea_surface_wind_wave_mean_period

	VMDR_WW [degree] Mean wind wave direction from sea_surface_wind_wave_from_direction
	VHM0_SW1 [m] Spectral significant primary swell wave height sea_surface_primary_swell_wave_significant_height
	VTM01_SW1 [s] Spectral moments (0,1) primary swell wave period sea_surface_primary_swell_wave_mean_period
	VMDR_SW1 [degree] Mean primary swell wave direction from sea_surface_primary_swell_wave_from_direction
	VHM0_SW2 [m] Spectral significant secondary swell wave height sea_surface_secondary_swell_wave_significant_height
	VTM01_SW2 [s] Spectral moments (0,1) secondary swell wave period sea_surface_secondary_swell_wave_mean_period
	VMDR_SW2 [degree] Mean secondary swell wave direction from sea_surface_secondary_swell_wave_from_direction

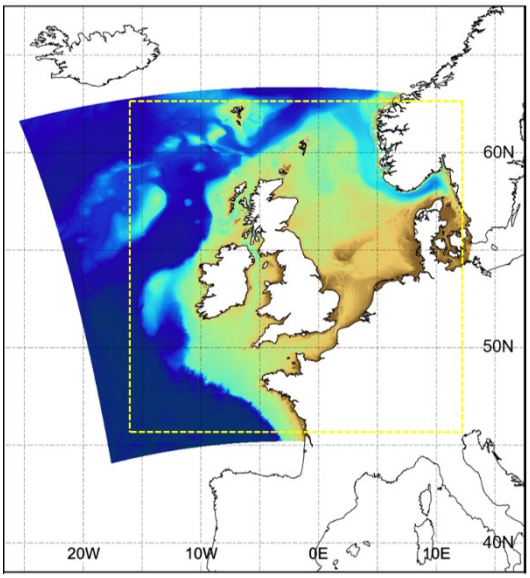
The model **bathymetry** and Land-Sea mask are stored in the static dataset:

cmems_mod_nws_wav_my_1.5km-3D_static	**Variable name** in the NetCDF files (unit) Long_name Standard_name
	mask Land-sea mask sea_binary_mask
	depth [m] depth depth
	deptho [m] Bathymetry sea_floor_depth_below_geoid

II.3 Product System Description

Products from the North-West European Shelf Wave Reanalysis system are generated using a WAVEWATCH III (version 4.18) 1.5 km Atlantic Margin Model (hereafter denoted as WWIII-AMM15). The wave model provides a description of ocean surface gravity wave (periods 3-30 seconds) characteristics as an extension to the existing physical and ecosystem model products provided by the

North-West Shelf MFC. The following subsections describe the model component and its dependencies in terms of models providing forcing and lateral boundary conditions.

Geographical domain/ Resolution	NORTH WEST EUROPEAN SHELF (16°W -13°E; 46°N – 63°N) 1/33° longitude x 1/74° latitude (958 x 1240 grid points) The products are covering the area highlighted by the yellow dashed line.
Geographic coverage	 *NWS bathymetry (m) showing in yellow the domain of the 004_015 product.* The domain extends beyond the continental shelf in order to place the model's boundary region in the deep waters of the adjacent North-East Atlantic, but the focus region for the model comprises open waters of the shelf seas, i.e. (using UK terminology) the North Sea, Irish Sea, English Channel, Celtic Sea and Bay of Biscay. The model is a nested regional wave model configuration. The domain extends beyond the continental shelf in order to place the model's boundary region in the deep waters of the adjacent North-East Atlantic, but the focus region for the model comprises open waters of the shelf seas, i.e. (using UK terminology) the North Sea, Irish Sea, English Channel, Celtic Sea and Bay of Biscay.

Horizontal grid	1.9± 0.4 km longitude x 1.5km latitude The longitude grid resolution varies with latitude. It is coarser at lower latitude (lat=46°N ->1,3 km), and finer at higher latitude (lat=63°N ->1.5km). The latitudinal grid resolution is constant in the whole model domain and is 1.5km). The wave model native horizontal grid is a Spherical Multiple Cell grid with a variable resolution of 3-1.5 km, designed for improved resolution of islands and coastlines. 3 km (1/30° x 1/30°) for water depths >40m; 1.5 km (1/60° x 1/60°) for water depths <= 40m.
Vertical grid	Wave models describe ocean surface conditions only, so do not require a vertical coordinate system. Furthermore, the majority of effects on wave propagation and dissipation occur in waters of intermediate or shallow depth as defined relative to the wave frequency (inverse of wave period): $d \leq \frac{g}{4\pi f^2}$; where d is water depth, g acceleration due to gravity and f is wave frequency. Thus, bathymetry is only a controlling mechanism on the wave field for depths below approximately 490 m, based on a minimum frequency in the model of approximately 0.04 Hz (period 25 seconds).
Spectral grid	WWIII-AMM15 wave spectra are defined on a two-dimensional frequency-direction grid using 24 (15°) direction cells and 30 geometrically spaced frequency cells. The geometric spacing is used in order to enable an efficient application of the model's Discrete Interaction Approximation parameterisation of nonlinear wave-wave interactions (Hasselmann et al., 1985) and is based on a multiplication factor of 1.1. The minimum frequency for the configuration is set at 0.4118 Hz, such that the maximum frequency bin is at 0.6532 Hz and the prognostic part of the wave model covers periods from approximately 25 to 1.5 seconds. In order to include the important contribution of higher frequency waves to wave growth/dissipation processes and the output wave characteristics a parametric tail is fitted for frequencies above the spectral maximum (e.g. WAMDIG, 1988), in this case using an f^{-5} relationship (Tolman, 2014).
Model version	WAVEWATCH III® (version 4.18)
Atmospheric forcings	ECMWF- ERA 5 winds
Assimilation scheme	Ø
Assimilated observations	Ø
Boundaries conditions	The WWIII-AMM15S model is nested one-way with lateral boundary conditions supplied from a Met Office global wave model hindcast, driven

	by the same ERA5 winds. The wave boundary data are supplied as wave spectra, which are interpolated from the global model's 25 km cell resolution onto the (predominantly) 3 km open waters grid cells found at the wave model outer boundaries.
Initial conditions	15 day spin up from calm sea state
Bathymetry	The WWIII-AMM15 configuration uses the same bathymetry as the equivalent NWS analysis forecast wave and physics products (see CMEMS-NWS-PUM-004-014 and CMEMS-NWS-PUM-004-013), which has been derived from the EMODnet 2015 bathymetry. In order to match the grid set-up in the hydrodynamic model, but improve the estimate of wave energy transmission in the vicinity of sub grid scale topographic features such as small islands and coastal headlands, a number of sub grid blocking cells are defined for the wave model following the method described by Chawla and Tolman (2008).

II.4 Processing information

II.4.1 Update Time

The times series is extended twice a year, adding 6 months.

II.4.2 Time coverage

From 01/01/1980 and regularly updated (see product improvements pages http://marine.copernicus.eu/services-portfolio/product-improvements/).

II.4.3 Time averaging

The fields are 3-hourly instantaneous at 00, 03, 06, 09, 12, 15, 18, 21 UTC.

II.4.4 Time variable in netCDF files

Time is expressed in seconds since epoch 1970-01-01, as follows:

```
double time(time) ;
time:units = "seconds since 1970-01-01T00:00:00Z" ;
time:long_name = "Validity time" ;
time:standard_name = "time" ;
time:calendar = "Gregorian" ;
```

```
time:axis = "T" ;
```

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this articles " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center"

IV.1 Nomenclature of files when downloaded through the Copernicus Marine Service Web Portal Service

The NORTHWESTSHELF_ANALYSIS_FORECAST_WAV_004_014 files nomenclature when downloaded through Marine Data Store is based on product dataset name and a numerical reference related to the request date on the CIS.

The scheme is: **datasetname-nnnnnnnnnnnnn.nc**

where:

- **datasetname:** as described in table 2
- **nnnnnnnnnnnnn**: 13 digit integer corresponding to the current time (download time) in milliseconds since January 1, 1970 midnight UTC.
- **.nc**: standard NetCDF filename extension.

Example:

MetO-NWS-WAV-hi-1303461772348.nc

IV.2 File size (netCDF4)

DATASET NAME	FILE NAME	DIMENSION [MB]
MetO-NWS-WAV-RAN	metoffice_wave_amm15_NWS_WAV_3hi{yyyymmdd}.nc	60
cmems_mod_nws_wav_my_1.5km-3D_static	NWS-MFC_004_015_static.nc	155

IV.3 Remember: scale_factor & add_offset / missing_value / land mask

Real_Value = (Display_Value X scale_factor) + add_offset

The missing value for this product is: -32767s

Land and sea-ice masks are equal to "_FillValue" (see variable attribute on NetCDF file).

IV.4 Reading Software

NetCDF data can be browsed and used through a number of software, like:

- ✓ ncBrowse: http://www.epic.noaa.gov/java/ncBrowse/,
- ✓ NetCDF Operator (NCO): http://nco.sourceforge.net/
- ✓ Panoply (NASA): https://www.giss.nasa.gov/tools/panoply/
- ✓ IDL, Matlab, GMT…

Useful information on UNIDATA: http://www.unidata.ucar.edu/software/netcdf/

IV.5 Structure of files

netcdf metoffice_wave_amm15_NWS_WAV_3hi20191231, see in ANNEX. {

V REFERENCES

Quality Information Document (QUID): https://documentation.marine.copernicus.eu/QUID/CMEMS-NWS-QUID-004-015.pdf

VI ANNEX

```
netcdf metoffice_wave_amm15_NWS_WAV_3hi20191231. {
dimensions:
        level = 1 ;
        longitude = 958 ;
        latitude = 1240 ;
        time = UNLIMITED ; // (8 currently)
variables:
        float longitude(longitude) ;
                longitude:units = "degree_east" ;
                longitude:longitudeg_name = "longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:axis = "X" ;
                longitude:valid_min = -180.f ;
                longitude:valid_max = 360.f ;
        float latitude(latitude) ;
                latitude:units = "degree_north" ;
                latitude:longitudeg_name = "latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:valid_min = -90.f ;
                latitude:valid_max = 90.f ;
                latitude:axis = "Y" ;
        int crs ;
                crs:grid_mapping_name = "latitude_longitude" ;
                crs:longitude_of_prime_meridian = 0.f ;
                crs:semi_major_axis = 6378137.f ;
                crs:inverse_flatitudetening = 298.2572f ;
        double time(time) ;
                time:longitudeg_name = "time" ;
                time:standard_name = "time" ;
                time:units = "seconds since 1970-01-01T00:00:00Z" ;
                time:calendar = "standard" ;
```

```
                time:axis = "T" ;
        short VHM0(time, latitude, longitude) ;
                VHM0:longitudeg_name = "Spectral significant wave height (Hm0)" ;
                VHM0:standard_name = "sea_surface_wave_significant_height" ;
                VHM0:units = "m" ;
                VHM0:_FillValue = -32767s ;
                VHM0:scale_factor = 0.002f ;
                VHM0:add_offset = 0.f ;
                VHM0:valid_min = 0 ;
                VHM0:valid_max = 16000 ;
                VHM0:cell_methods = "time:point" ;
                VHM0:grid_mapping = "crs" ;
        short VTM02(time, latitude, longitude) ;
                VTM02:longitudeg_name = "Spectral moments (0,2) wave period Tm02" ;
                VTM02:standard_name =
"sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_moment" ;
                VTM02:units = "s" ;
                VTM02:_FillValue = -32767s ;
                VTM02:scale_factor = 0.01f ;
                VTM02:add_offset = 0.f ;
                VTM02:valid_min = 0 ;
                VTM02:valid_max = 3000 ;
                VTM02:cell_methods = "time:point" ;
                VTM02:grid_mapping = "crs" ;
        short VTM10(time, latitude, longitude) ;
                VTM10:longitudeg_name = "Spectral moments (-1,0) wave period (Tm-10)" ;
                VTM10:standard_name =
"sea_surface_wave_mean_period_from_variance_spectral_density_inverse_frequency_moment" ;
                VTM10:units = "s" ;
                VTM10:_FillValue = -32767s ;
                VTM10:scale_factor = 0.01f ;
                VTM10:add_offset = 0.f ;
                VTM10:valid_min = 0 ;
                VTM10:valid_max = 3000 ;
```

```
                VTM10:cell_methods = "time:point" ;
                VTM10:grid_mapping = "crs" ;
        short VTPK(time, latitude, longitude) ;
                VTPK:longitudeg_name = "Wave period at spectral peak / peak period (Tp)" ;
                VTPK:standard_name =
"sea_surface_wave_period_at_variance_spectral_density_maximum" ;
                VTPK:units = "s" ;
                VTPK:_FillValue = -32767s ;
                VTPK:scale_factor = 0.01f ;
                VTPK:add_offset = 0.f ;
                VTPK:valid_min = 0 ;
                VTPK:valid_max = 3000 ;
                VTPK:cell_methods = "time:point" ;
                VTPK:grid_mapping = "crs" ;
        short VMDR(time, latitude, longitude) ;
                VMDR:longitudeg_name = "Mean wave direction from (Mdir)" ;
                VMDR:standard_name = "sea_surface_wave_from_direction" ;
                VMDR:units = "degree" ;
                VMDR:_FillValue = -32767s ;
                VMDR:scale_factor = 0.1f ;
                VMDR:add_offset = 0.f ;
                VMDR:valid_min = 0 ;
                VMDR:valid_max = 3600 ;
                VMDR:direction_reference = "True North" ;
                VMDR:cell_methods = "time:point" ;
                VMDR:grid_mapping = "crs" ;
        short VPED(time, latitude, longitude) ;
                VPED:longitudeg_name = "Wave principal direction at spectral peak" ;
                VPED:standard_name =
"sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
                VPED:units = "degree" ;
                VPED:_FillValue = -32767s ;
                VPED:scale_factor = 0.1f ;
                VPED:add_offset = 0.f ;
```

VPED:valid_min = 0 ;

VPED:valid_max = 3600 ;

VPED:direction_reference = "True North" ;

VPED:cell_methods = "time:point" ;

VPED:grid_mapping = "crs" ;

short VHM0_WW(time, latitude, longitude) ;

VHM0_WW:longitudeg_name = "Spectral significant wind wave height" ;

VHM0_WW:standard_name = "sea_surface_wind_wave_significant_height" ;

VHM0_WW:units = "m" ;

VHM0_WW:_FillValue = -32767s ;

VHM0_WW:scale_factor = 0.002f ;

VHM0_WW:add_offset = 0.f ;

VHM0_WW:valid_min = 0 ;

VHM0_WW:valid_max = 16000 ;

VHM0_WW:comment = "Wind sea and swells defined using topographic partitions and spectral wave-age cut-off" ;

VHM0_WW:cell_methods = "time:point" ;

VHM0_WW:grid_mapping = "crs" ;

short VHM0_SW1(time, latitude, longitude) ;

VHM0_SW1:longitudeg_name = "Spectral significant primary swell wave height" ;

VHM0_SW1:standard_name = "sea_surface_primary_swell_wave_significant_height" ;

VHM0_SW1:units = "m" ;

VHM0_SW1:_FillValue = -32767s ;

VHM0_SW1:scale_factor = 0.002f ;

VHM0_SW1:add_offset = 0.f ;

VHM0_SW1:valid_min = 0 ;

VHM0_SW1:valid_max = 16000 ;

VHM0_SW1:comment = "Wind sea and swells defined using topographic partitions and spectral wave-age cut-off" ;

VHM0_SW1:cell_methods = "time:point" ;

VHM0_SW1:grid_mapping = "crs" ;

short VHM0_SW2(time, latitude, longitude) ;

VHM0_SW2:longitudeg_name = "Spectral significant secondary swell wave height" ;

VHM0_SW2:standard_name = "sea_surface_secondary_swell_wave_significant_height" ;

VHM0_SW2:units = "m" ;

VHM0_SW2:_FillValue = -32767s ;

VHM0_SW2:scale_factor = 0.002f ;

VHM0_SW2:add_offset = 0.f ;

VHM0_SW2:valid_min = 0 ;

VHM0_SW2:valid_max = 16000 ;

VHM0_SW2:comment = "Wind sea and swells defined using topographic partitions and spectral wave-age cut-off" ;

VHM0_SW2:cell_methods = "time:point" ;

VHM0_SW2:grid_mapping = "crs" ;

short VMDR_WW(time, latitude, longitude) ;

VMDR_WW:longitudeg_name = "Mean wind wave direction from" ;

VMDR_WW:standard_name = "sea_surface_wind_wave_from_direction" ;

VMDR_WW:units = "degree" ;

VMDR_WW:_FillValue = -32767s ;

VMDR_WW:scale_factor = 0.1f ;

VMDR_WW:add_offset = 0.f ;

VMDR_WW:valid_min = 0 ;

VMDR_WW:valid_max = 3600 ;

VMDR_WW:comment = "Wind sea and swells defined using topographic partitions and spectral wave-age cut-off" ;

VMDR_WW:direction_reference = "True North" ;

VMDR_WW:cell_methods = "time:point" ;

VMDR_WW:grid_mapping = "crs" ;

short VMDR_SW1(time, latitude, longitude) ;

VMDR_SW1:longitudeg_name = "Mean primary swell wave direction from" ;

VMDR_SW1:standard_name = "sea_surface_primary_swell_wave_from_direction" ;

VMDR_SW1:units = "degree" ;

VMDR_SW1:_FillValue = -32767s ;

VMDR_SW1:scale_factor = 0.1f ;

VMDR_SW1:add_offset = 0.f ;

VMDR_SW1:valid_min = 0 ;

VMDR_SW1:valid_max = 3600 ;

VMDR_SW1:comment = "Wind sea and swells defined using topographic partitions and spectral wave-age cut-off" ;

VMDR_SW1:direction_reference = "True North" ;

VMDR_SW1:cell_methods = "time:point" ;

VMDR_SW1:grid_mapping = "crs" ;

short VMDR_SW2(time, latitude, longitude) ;

VMDR_SW2:longitudeg_name = "Mean secondary swell wave direction from" ;

VMDR_SW2:standard_name = "sea_surface_secondary_swell_wave_from_direction" ;

VMDR_SW2:units = "degree" ;

VMDR_SW2:_FillValue = -32767s ;

VMDR_SW2:scale_factor = 0.1f ;

VMDR_SW2:add_offset = 0.f ;

VMDR_SW2:valid_min = 0 ;

VMDR_SW2:valid_max = 3600 ;

VMDR_SW2:comment = "Wind sea and swells defined using topographic partitions and spectral wave-age cut-off" ;

VMDR_SW2:direction_reference = "True North" ;

VMDR_SW2:cell_methods = "time:point" ;

VMDR_SW2:grid_mapping = "crs" ;

short VTM01_WW(time, latitude, longitude) ;

VTM01_WW:longitudeg_name = "Spectral moments (0,1) wind wave period" ;

VTM01_WW:standard_name = "sea_surface_wind_wave_mean_period" ;

VTM01_WW:units = "s" ;

VTM01_WW:_FillValue = -32767s ;

VTM01_WW:scale_factor = 0.01f ;

VTM01_WW:add_offset = 0.f ;

VTM01_WW:valid_min = 0 ;

VTM01_WW:valid_max = 3000 ;

VTM01_WW:comment = "Wind sea and swells defined using topographic partitions and spectral wave-age cut-off" ;

VTM01_WW:cell_methods = "time:point" ;

VTM01_WW:grid_mapping = "crs" ;

short VTM01_SW1(time, latitude, longitude) ;

VTM01_SW1:longitudeg_name = "Spectral moments (0,1) primary swell wave period" ;

VTM01_SW1:standard_name = "sea_surface_primary_swell_wave_mean_period" ;

VTM01_SW1:units = "s" ;

VTM01_SW1:_FillValue = -32767s ;

VTM01_SW1:scale_factor = 0.01f ;

VTM01_SW1:add_offset = 0.f ;

VTM01_SW1:valid_min = 0 ;

VTM01_SW1:valid_max = 3000 ;

VTM01_SW1:comment = "Wind sea and swells defined using topographic partitions and spectral wave-age cut-off" ;

VTM01_SW1:cell_methods = "time:point" ;

VTM01_SW1:grid_mapping = "crs" ;

short VTM01_SW2(time, latitude, longitude) ;

VTM01_SW2:longitudeg_name = "Spectral moments (0,1) secondary swell wave period" ;

VTM01_SW2:standard_name = "sea_surface_secondary_swell_wave_mean_period" ;

VTM01_SW2:units = "s" ;

VTM01_SW2:_FillValue = -32767s ;

VTM01_SW2:scale_factor = 0.01f ;

VTM01_SW2:add_offset = 0.f ;

VTM01_SW2:valid_min = 0 ;

VTM01_SW2:valid_max = 3000 ;

VTM01_SW2:comment = "Wind sea and swells defined using topographic partitions and spectral wave-age cut-off" ;

VTM01_SW2:cell_methods = "time:point" ;

VTM01_SW2:grid_mapping = "crs" ;

short VSDX(time, latitude, longitude) ;

VSDX:longitudeg_name = "Stokes drift U" ;

VSDX:standard_name = "sea_surface_wave_stokes_drift_x_velocity" ;

VSDX:units = "m s-1" ;

VSDX:_FillValue = -32767s ;

VSDX:scale_factor = 0.0005f ;

VSDX:add_offset = 0.f ;

VSDX:valid_min = -10000 ;

```
                VSDX:valid_max = 10000 ;
                VSDX:cell_methods = "time:point" ;
                VSDX:grid_mapping = "crs" ;
        short VSDY(time, latitude, longitude) ;
                VSDY:longitudeg_name = "Stokes drift V" ;
                VSDY:standard_name = "sea_surface_wave_stokes_drift_y_velocity" ;
                VSDY:units = "m s-1" ;
                VSDY:_FillValue = -32767s ;
                VSDY:scale_factor = 0.0005f ;
                VSDY:add_offset = 0.f ;
                VSDY:valid_min = -10000 ;
                VSDY:valid_max = 10000 ;
                VSDY:cell_methods = "time:point" ;
                VSDY:grid_mapping = "crs" ;

// global attributes:
                :Conventions = "CF-1.7" ;
                :institution = "UK Met Office" ;
                :contact = "servicedesk.cmems@mercator-ocean.eu" ;
                :references = "http://marine.copernicus.eu/" ;
                :title = "3-hourly-instantaneous wave" ;
                :source = "Met Office Operational Suite, Atlantic Margin Model (AMM15S 3-1.5km) WAVEWATCH III" ;
                :forcing_data_source = "ECMWF; ERA-5\"" ;
                :netcdf-version-id = "NETCDF4" ;
                :grid_id = "wave_nwshelf_standard_v1.1" ;
                :history = "File created: 2020-02-28T20:38:32Z" ;
}
```